Investigation of Sionna RT for Ray-Traced ISAC-Based Sensing

Design project presentation

Nandish Chauhan | Aahan Poddar

Indian Institute of Information Technology Vadodara

Under the guidance of Dr. Bhupendra Kumar

Why study radio propagation?

- A transmitted signal reaches the receiver through multiple paths.
- Buildings and surfaces create reflections, diffraction, and blockage.
- These paths affect communication quality and carry information about the environment.

Key question

Can a ray tracer generate useful, interpretable path data for communication analysis and future sensing studies?

ISAC and the scope of this project

ISAC motivation

Integrated sensing and communication seeks to use radio signals and resources for both connectivity and sensing.

What we completed

A moving-receiver ray-tracing simulation and analysis of path geometry, power, delay, and Doppler.

Project boundary: No target detector, sensing waveform, range–Doppler processor, or sensing accuracy evaluation has been implemented. These are future uses of the outputs.

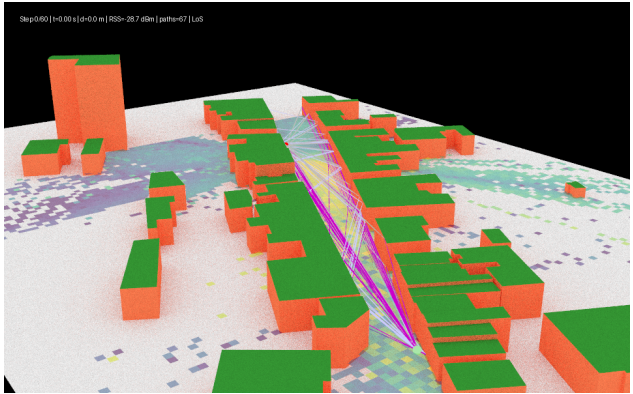
End-to-end project workflow

1. Load the 3D scene and configure materials, antennas, and frequency.
2. Define the receiver's waypoint route and sample its positions.
3. Compute a static radio map and trace paths at each snapshot.
4. Extract complex coefficients, delay, Doppler, angles, and interactions.
5. Calculate channel metrics and export datasets and visualizations.

Main implementation

`mobility_channel_study.ipynb` using Sionna RT 2.1.0.

3D scene and modeled environment



- Imported scene: building meshes and ground plane.
- Materials: ITU-style marble, metal, and concrete.
- Fixed transmitter; receiver travels through the scene.
- Colored lines show traced paths; the overlay shows the static radio map.

Simulation configuration

Parameter	Saved run
Frequency / bandwidth	3.5 GHz / 20 MHz
Transmitter power	44 dBm
TX / RX arrays	4×4 / 1×1
TX position	$(-107, 43, 10)$ m
Receiver height / nominal speed	2 m / 10 m/s
Route / snapshots	1183.45 m / 61
Path depth / samples	3 / 100,000
Radio map	400 m $\times$ 400 m; 4 m cells

Balanced profile

A practical exploratory configuration; not a convergence or field-validation study.

Run: `mobility_20261104_235027`

What propagation paths are modeled?

- **LoS:** direct path between transmitter and receiver.
- **Specular reflection:** path reflected from a surface.
- **Refraction:** transmission through a modeled material.
- **Diffraction:** path involving an edge or obstruction.
- **Mixed:** a path contains multiple interaction types.
- Diffuse reflection is enabled, but scattering coefficients are zero; no diffuse-category records appear.

How the notebook performs ray tracing

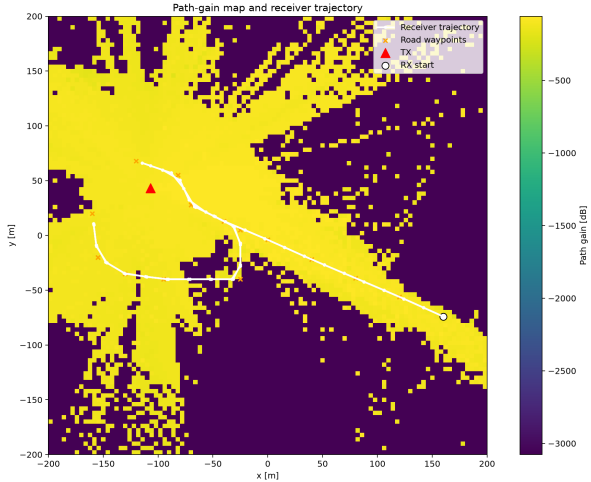
Core sequence (simplified)

```
scene = load_scene(SCENE_PATH)
scene.frequency = 3.5e9
scene.add(tx); scene.add(rx)

for step in range(num_steps):
    rx.position = trajectory[step]
    rx.velocity = velocity[step]
    paths = path_solver(scene, ...)
    a, tau = paths.cir(
        normalize_delays=False,
        out_type="numpy")
```

- Receiver position and velocity change at each snapshot.
- The solver retraces valid paths for that position.
- Channel impulse response data and path metadata are stored.
- Radio map is computed separately.

Receiver route and radio map



- Waypoints describe an out-and-back route with side-road branches.
- Route length: **1183.45 m**; duration: **118.35 s** at nominal speed.
- 61 sampled positions; approximately 19.72 m apart along the route.
- Map height is 1.5 m; receiver height is 2 m. Map gain is not the same quantity as snapshot RSS.

From traced paths to channel metrics

For each path, equal-weight transmitter-array combining gives an effective complex coefficient h_i.

Coherent received power

$$P_{\mathrm{coh}} = P_t \left| \sum_i h_i \right|^2$$

Includes constructive and destructive interference.

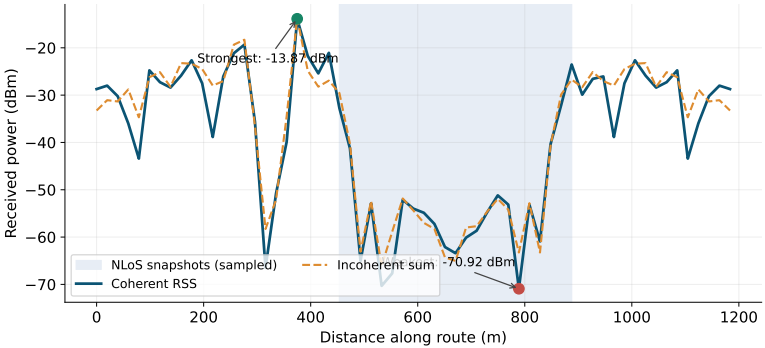
Incoherent path-power sum

$$P_{\mathrm{inc}} = P_t \sum_i |h_i|^2$$

Adds path powers without inter-path phase interference.

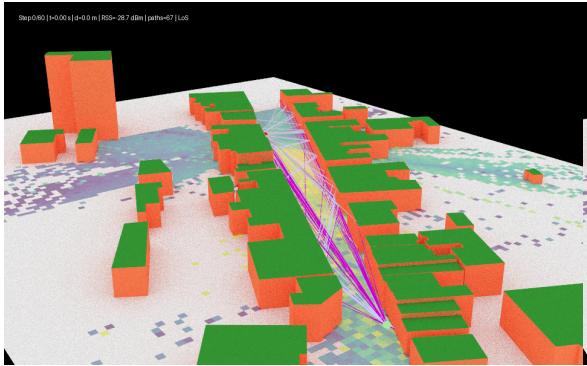
The notebook also calculates power-weighted RMS delay and Doppler spreads, path counts, and thermal-noise-only SNR.

Received power along the route

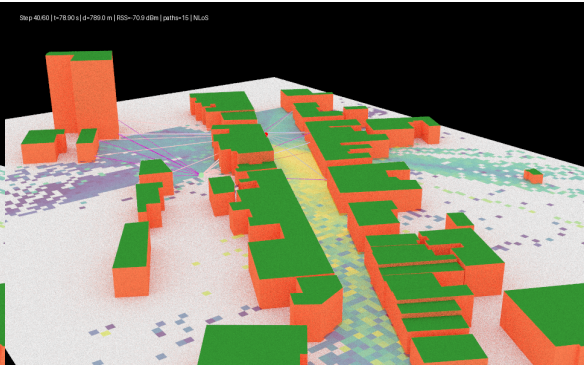


- Strongest: -13.87 **dBm**, step 19.
- Weakest: -70.92 **dBm**, step 40.
- Coherent and incoherent curves differ because of path phase.
- Shading marks sampled NLoS positions, not exact blockage boundaries.

LoS blockage and reappearance



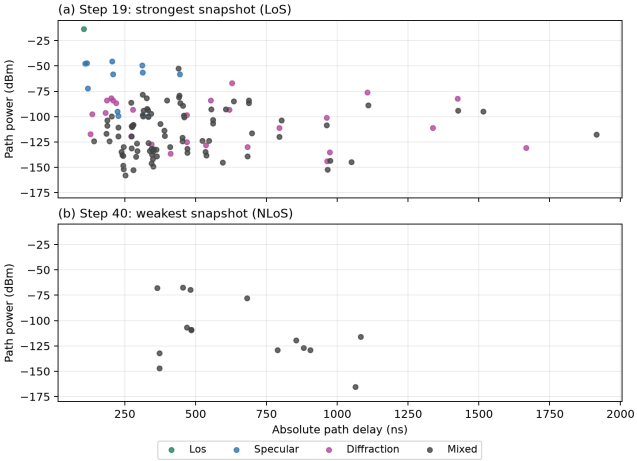
Step 0: LoS, 67 valid paths



Step 40: NLoS, 15 valid paths

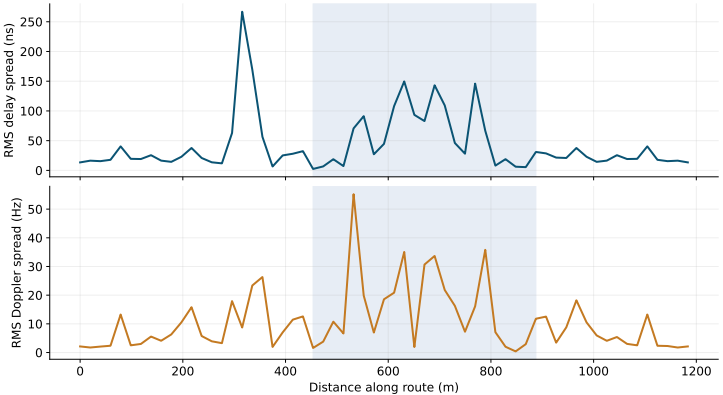
- **39 LoS** and **22 NLoS** snapshots; LoS is absent at steps 23–44.
- All 61 snapshots have at least one valid traced path. NLoS is not automatically a service outage.

Path delays and propagation mechanisms



- **4,826** valid path records across the run.
- Mixed: 3,426; diffraction-only: 1,010; specular-only: 337.
- LoS: 39; refraction-only: 14; diffuse-only: 0.
- Each point is one path's power and absolute delay.
- Counts do not represent power shares.

Delay and Doppler spread



- RMS delay spread: **2.40–266.86 ns**.
- RMS Doppler spread: **0.39–55.21 Hz**.
- Values are power-weighted across traced paths.
- Doppler depends on the velocity used by the snapshot model.

What does the code export?

Numerical data

`path_data.csv`: one record per path.

`snapshot_summary.csv`: aggregate metrics.

`channel_arrays.npz`: numerical channel arrays and validity mask.

`run_config.json`: saved run settings and material audit.

Visual outputs

Radio map and trajectory; channel metrics; path geometry; power–delay profile; scene animation.

Fields include delay, Doppler, angles, complex coefficient, interaction type, material, and vertices.

How could these outputs support ISAC?

- **Delay:** build channel impulse responses and study path separability.
- **Doppler:** investigate signatures related to modeled relative motion.
- **Angles and vertices:** relate multipath components to scene geometry.
- **Complex coefficients:** create frequency-selective channels for downstream communication or sensing experiments.

Important distinction

These are one-way transmitter–receiver paths. They are not automatically radar target ranges; target geometry, a waveform, processing, and validation must be added.

Limitations and next steps

Current limitations

- One simulated route and static scene; no field measurements.
- Material properties are not calibrated.
- No moving targets, interference, or receiver noise figure.
- Chord-based velocity can underestimate speed near sharp turns.

Next steps

- Test sensitivity to path samples, depth, and snapshot spacing.
- Refine trajectory and calibrate materials.
- Add a defined sensing target and waveform.
- Implement and validate range, Doppler, or angle estimation.

Conclusion and references

Takeaway

Sionna RT provides interpretable, geometry-based channel data. Our project demonstrates the ray-tracing and analysis pipeline that can serve as a foundation for later ISAC sensing studies.

References

F. Aït Aoudia *et al.*, "Sionna RT: Technical Report," arXiv:2504.21719, 2025.

F. Liu *et al.*, "Integrated sensing and communications: Toward dual-functional wireless networks for 6G and beyond," *IEEE JSAC*, 2022.

Thank you — Questions?